

Accuracy Is Not Adoption in AI Assisted eReferral Triage

Douglas KAVANAGH, BSE, MD, CCFP
OceanMD, Toronto, Ontario, Canada

Corresponding author: Douglas Kavanagh, dkavanagh@oceanmd.com

Abstract. Health systems often evaluate clinical artificial intelligence by accuracy and latency, although sustained adoption also depends on whether operational value exceeds implementation burden. We report a single-site implementation of an artificial intelligence-assisted eReferral triage tool in an outpatient cardiology clinic. After a four-month refinement period, system logs and clinic feedback were reviewed for a six-week demonstration. The clinic received 87 referrals; 33 (38%) triggered an alert, including five referrals marked routine by the sender that met the clinic's urgent-review criteria. Mean automated time to action was 24.34 seconds, and no manual overrides were recorded. The clinic nevertheless reported limited overall workflow impact and later discontinued use because low referral volume did not justify setup, oversight, and ongoing technical maintenance. The chest-pain rule also produced clinically mixed matches, despite aggregate performance metrics showing no recorded outliers. This case suggests that technical performance is necessary but insufficient for adoption. Health systems should assess minimum viable referral volume, workflow readiness, maintenance ownership, human oversight, and total implementation effort before launch. High-volume central intake programs are likely better settings for evaluating whether AI-assisted triage produces sustainable operational benefit.

Keywords. artificial intelligence, eReferral, clinical workflow, implementation, sustainability

1. Introduction

Artificial intelligence can classify referral content and trigger workflow actions within seconds. These capabilities are relevant to referral intake, where staff interpret variable clinical narratives while managing queues and interruptions. Technical capability alone does not establish value. A tool must address enough work to justify configuration, monitoring, governance, and maintenance after a demonstration ends.

Health artificial intelligence guidance emphasizes oversight, transparency, accountability, and evaluation in the intended setting [1]. DECIDE-AI calls for early clinical evaluations to report workflow, human factors, safety, and performance rather than accuracy alone [2]. The NASSS framework explains why promising technologies may be abandoned when complexity affects their value proposition, organization, wider system, or adaptation over time [3]. This paper examines a live implementation that generated rapid alerts but had limited workflow impact and was not sustained.

2. Setting and Intervention

The demonstration took place at Cardiac Care North York, an outpatient cardiology clinic using the Ocean electronic referral platform. Eleven clinic users participated in referral intake workflows. Thirty-six unique senders submitted referrals during the demonstration period. The participating site was the only receiving clinic connected to the automated routing interface in the final project scope.

The Autorouter was a separate application that subscribed to eReferral events, analyzed structured and narrative referral content using a site-selected large language model, evaluated plain-language rules, and called bounded workflow tools. The implementation focused on two alerting rules rather than comprehensive autonomous routing. One rule identified referrals that might require urgent review based on criteria such as new exertional chest pain, acute coronary syndrome, exertional syncope, or critical valvular disease. The second identified referrals that might be suitable for the clinic's Chest Pain Clinic.

When a rule matched, the system emailed an alert and recommendation to booking coordinators, who retained responsibility for review and action. An earlier auto-acceptance function was disabled because it bypassed the point when staff created an electronic medical record entry. The application therefore functioned as a screening and notification layer rather than an autonomous triage service.

3. Evaluation Approach

This descriptive implementation report draws on the final Canada Health Infoway Vendor Innovation Program demonstration report, system activity logs, aggregate referral analytics, and clinic feedback [4]. The project included a refinement period from June 24 to October 13, 2025. OceanMD and clinic staff reviewed rule-triggered referrals, investigated unexpected behavior, tested simulated cases, and revised prompts and rule instructions.

The prospective demonstration ran from October 13 to November 24, 2025. Measures included referral and trigger counts, recorded outliers, action-execution success, manual overrides, and elapsed time from submission to the automated action. A non-concurrent six-week baseline supplied referral volume and estimated workflow timing. The source report did not document independent clinical adjudication or structured review of every non-triggered referral. We therefore report observed alerts and recorded outliers rather than interpreting the reported 100% rule-match accuracy as validated diagnostic performance. The operational evaluation used aggregate data and staff feedback; no patient-identifying data are reported.

4. Results

The clinic received 87 referrals during the six-week demonstration, compared with 76 referrals during the earlier six-week baseline period. Thirty-three of the 87 demonstration referrals (38%) triggered at least one rule. The system generated its routing-related action in a mean of 24.34 seconds for triggered referrals.

Five referrals labelled routine by the sending provider met the clinic-defined urgent-review criteria and generated alerts. Three referrals labelled urgent by the sender also generated urgent alerts. These observations show that referral narrative contained signals that were not always represented by the sender-assigned priority.

Activity records reported no manual overrides, missed actions, or incorrect action executions among 33 triggered referrals. These findings followed four months of refinement and indicate stable execution for a narrow configuration during a short observation window. They do not establish perfect sensitivity or specificity.

Clinic feedback qualified the aggregate metrics. Staff described the urgent-referral rule as highly accurate when physician notes explicitly contained words such as urgent or priority. They described mixed results for the Chest Pain Clinic rule because general symptom descriptions could mention chest pain without meeting the program's intended criteria. The clinic's overall assessment was that the system had limited impact on its referral workflow at that stage.

Booking coordinators estimated that manual review required approximately two minutes per referral and did not routinely process referrals on weekends. Alerts could therefore provide earlier awareness, but the evaluation did not measure actual staff time saved, response after alerting, changes in management, or patient outcomes.

The clinic did not sustain use after the demonstration. Follow-up attributed discontinuation to low referral volume relative to setup, oversight, and ongoing technical maintenance. The limited number of actionable alerts did not create enough operational benefit to justify continued maintenance. No safety failure was identified as the reason for discontinuation.

5. Discussion

This case separates technical performance from adoption. The application processed referrals quickly, executed configured actions reliably, and identified several referrals that warranted earlier attention. Those results were insufficient to sustain use because the total amount of work addressed was small. Thirty-three alerts across six weeks represented fewer than one triggered referral per working day. Even a useful alert may not offset the fixed effort required to document workflow, configure rules, review behavior, maintain integrations, and assign ongoing ownership.

Referral volume affects both potential benefit and unit economics. The time saved and number of important cases identified may increase with eligible referrals, while configuration, privacy review, training, monitoring, and maintenance remain fixed or stepwise. Low-volume sites may therefore face an unfavorable benefit-to-burden ratio despite acceptable performance. Volume is not sufficient on its own. The abandoned auto-acceptance function shows that automation must account for adjacent tasks and exception paths. Workflow discovery should precede configuration, with human review preserved until the surrounding process is understood.

The project also identified a deployment-selection problem. Wider use has been constrained by limited awareness, variable organizational readiness, and difficulty engaging central intake programs with enough electronic referral activity to justify implementation. These explanations reflect project-team experience rather than formally collected comparative data. They should be tested in future implementations instead of treated as established causal findings.

High-volume central intake programs are a logical next setting because they concentrate referrals, staff expertise, routing rules, and accountability. They may provide enough recurring work to justify governance and maintenance while producing sufficient cases for prospective validation. This is a testable hypothesis, not a demonstrated scaling outcome.

6. Recommendations

Health systems and funders should add a volume-and-readiness assessment before selecting sites for artificial intelligence workflow pilots. The assessment should estimate eligible referral volume, current handling time, frequency of actionable cases, fixed implementation effort, expected maintenance, and the consequence of missed or incorrect actions. A willing site is not necessarily an informative demonstration environment.

Implementation should begin with a small number of advisory rules. Teams should document adjacent work before enabling automation. Each rule needs an accountable owner, review process, exception pathway, and pause or retirement criteria. Prospective evaluation should review triggered and non-triggered referrals against an appropriate standard and distinguish automated action from human review and clinical resolution.

Evaluations should measure actual staff effort, alert response, overrides, false positive and false negative results, workflow effects, maintenance time, and willingness to continue. Innovation programs should connect automation projects with settings that have enough digital activity to test the value proposition. As standards-based eReferral adoption expands [5], central intake programs may offer the volume and governance required to evaluate durable benefit.

7. Limitations

This report describes one clinic, one narrow configuration, 87 referrals, and six weeks of prospective observation. The baseline was non-concurrent, review time was estimated, and no labor savings, patient outcomes, response after alerting, or total cost were measured. No independent adjudication or systematic review of all non-triggered referrals was documented, and clinic feedback identified mixed Chest Pain Clinic matches. Proposed reasons for limited wider deployment were not evaluated through structured interviews or site comparisons. The relationship between higher volume and sustainability remains a hypothesis.

8. Conclusion

A live clinical implementation of artificial intelligence-assisted referral screening generated rapid alerts and identified five sender-labelled routine referrals that met urgent-review criteria. The clinic nevertheless experienced limited overall impact and discontinued use when low referral volume did not justify implementation and technical maintenance effort. The case supports a practical policy conclusion: clinical artificial

intelligence pilots should be selected and evaluated according to workflow fit, addressable volume, human oversight, and the full burden of sustaining the service. Technical performance is one condition for adoption, not its endpoint.

Acknowledgments

This implementation was completed through Canada Health Infoway's Vendor Innovation Program with participation from Cardiac Care North York and project support from OceanMD staff.

Competing Interests

The author is Medical Director and Co-Founder of OceanMD, which developed the Autorouter reference implementation and operates the Ocean eReferral platform.

References

- [1] World Health Organization. Ethics and governance of artificial intelligence for health. Geneva: World Health Organization; 2021. ISBN 978-92-4-002920-0.
- [2] Vasey B, Nagendran M, Campbell B, et al. Reporting guideline for the early-stage clinical evaluation of decision support systems driven by artificial intelligence: DECIDE-AI. *Nat Med*. 2022;28:924-933. doi:10.1038/s41591-022-01772-9.
- [3] Greenhalgh T, Wherton J, Papoutsi C, et al. Beyond adoption: A new framework for theorizing and evaluating nonadoption, abandonment, and challenges to the scale-up, spread, and sustainability of health and care technologies. *J Med Internet Res*. 2017;19(11):e367. doi:10.2196/jmir.8775.
- [4] OceanMD. Enhancing the Ocean Provider Network: Leveraging Standards-Based Interoperability to Enable Advanced Clinical Decision Support and AI-Powered Tools for Provincial eReferral Systems. Canada Health Infoway Vendor Innovation Program Demonstration Report. Version 4.0. March 30, 2026.
- [5] Canada Health Infoway. Pan-Canadian eReferral/eConsult Specification. Available from: <https://accelero.infoway-inforoute.ca/en/initiatives/ereferral-econsult>. Accessed September 7, 2026.